

National Technical University of Athens

School of Electrical and Computer Engineering

Modeling Smart Electric Vehicle Charging

**A Multi-Level Intelligence Approach with
Renewable Energy Integration**

Diploma Thesis

by

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II

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Vehicle-to-Grid (V2G)...
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Abstract

[Write your abstract in English - 300-500 words]

This diploma thesis investigates the application of smart electric vehicle (EV) charging systems in office buildings with renewable energy integration. Different levels of intelligence are analyzed, ranging from simple rule-based approaches to advanced optimization algorithms and Vehicle-to-Grid (V2G) technology...

Keywords: Electric vehicles, smart charging, renewable energy, optimization, Vehicle-to-Grid, office buildings

Acknowledgments

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Contents

II	1
Abstract	2
Acknowledgments	3
1 Introduction	12
1.1 Background and Motivation	12
1.2 Smart Charging Levels	12
1.2.1 Level 0: Baseline (Uncontrolled Charging)	12
1.2.2 Level 1: Simple Rule-Based Charging	13
1.2.3 Level 2: Intelligent Optimization	13
1.2.4 Level 3: Vehicle-to-Grid (V2G)	13
1.3 Role of Renewable Energy	14
1.4 Benefits of Smart Charging in Office Buildings	14
1.5 Literature Review	14
1.5.1 Smart EV Charging	14
1.5.2 Optimization Algorithms	15
1.5.3 Vehicle-to-Grid Technology	15
1.5.4 Renewable Integration	15
1.6 Research Objectives	15
1.7 Thesis Structure	15
2 Methodology	17
2.1 System Components	17
2.1.1 Office Building Model	17
2.1.2 Electric Vehicle Fleet Model	19
2.1.3 Grid Model	20
2.2 Optimization Problem Formulation	21
2.2.1 Objective Function	21
2.2.2 Energy Balance	21

2.3	Level 0: Baseline (Uncontrolled Charging)	21
2.3.1	Algorithm Description	21
2.3.2	Mathematical Formulation	22
2.4	Level 1: Simple Rule-Based Charging	22
2.4.1	Algorithm Description	22
2.4.2	Mathematical Formulation	22
2.5	Level 2: Intelligent Optimization Algorithms	23
2.5.1	Reinforcement Learning (Q-Learning)	23
2.6	Level 3: Vehicle-to-Grid (V2G)	24
2.6.1	Bidirectional Energy Flow	24
2.6.2	V2G Constraints	24
2.6.3	Revenue Calculation	24
2.6.4	V2G Integration with Algorithms	24
2.7	Key Assumptions and Limitations	25
2.7.1	Assumptions	25
2.7.2	Limitations	25
3	Case Study: Office Building Scenario	26
3.1	Building Energy Profile	26
3.1.1	Energy Consumption Data	26
3.1.2	Solar PV System	28
3.1.3	Building Battery Storage	29
3.2	Electric Vehicle Fleet	30
3.2.1	Fleet Composition	30
3.2.2	Mobility Patterns	31
3.3	Grid Parameters	32
3.3.1	Electricity Pricing	32
3.3.2	Grid Capacity Constraints	33
3.4	Simulation Configuration	34
3.4.1	Time Parameters	34
3.4.2	Algorithm Parameters	34
3.5	Data Sources Summary	35
3.6	Simulation Scenarios	35
3.6.1	Scenario 1: Charge-Only (No V2G)	35
3.6.2	Scenario 2: V2G-Enabled	35
3.7	Evaluation Metrics	36
3.7.1	Economic Metrics	36
3.7.2	Energy Metrics	36
3.7.3	Performance Metrics	36

3.7.4	Computational Metrics	36
3.8	Implementation	37
3.8.1	Software Environment	37
3.8.2	Code Structure	37
4	Results and Analysis	38
4.1	Overview of Results	38
4.2	Scenario 1: Charge-Only Results	38
4.2.1	Economic Performance	38
4.2.2	Energy Performance	39
4.2.3	Operational Performance	39
4.3	Scenario 2: V2G-Enabled Results	40
4.3.1	Economic Performance	40
4.3.2	V2G Discharge Patterns	40
4.3.3	Energy Flow Analysis	41
4.4	Comparative Analysis Across Intelligence Levels	42
4.4.1	Economic Comparison	42
4.4.2	Algorithm Performance Comparison	42
4.4.3	State of Charge Analysis	42
4.4.4	Grid Constraint Management	43
4.5	V2G Impact Assessment	43
4.5.1	Charge-Only vs V2G Comparison	43
4.5.2	V2G Revenue Breakdown	43
4.5.3	V2G Trade-offs	43
4.6	Sensitivity Analysis	44
4.6.1	Electricity Price Variation	44
4.6.2	Fleet Size Scaling	44
4.7	Algorithm-Specific Insights	45
4.7.1	Reinforcement Learning	45
4.8	Renewable Energy Integration	45
4.8.1	Solar Self-Consumption	45
4.9	Economic Analysis	45
4.9.1	Annual Projections	45
4.9.2	Return on Investment	45
5	Conclusions and Future Work	47
5.1	Summary of Findings	47
5.1.1	Intelligence Level Comparison	47
5.2	Answering the Research Questions	48
5.2.1	Which Intelligence Level Provides Substantial Benefits?	48

5.2.2	Is V2G Worth the Investment?	48
5.3	Challenges and Limitations	49
5.3.1	Technical Challenges	49
5.3.2	Economic Challenges	49
5.3.3	User Acceptance Challenges	49
5.3.4	Study Limitations	50
5.4	Future Research Directions	50
5.4.1	Short-Term Research Priorities	50
5.4.2	Medium-Term Research Directions	50
5.4.3	Long-Term Research Vision	51
5.4.4	Emerging Technologies	51
5.4.5	Environmental Impact	51
5.4.6	Social Impact	51
5.5	Final Remarks	51
5.5.1	Closing Statement	51
References		53
A Code Implementation		55
A.1	Project Structure	55
A.2	Core Models	56
A.2.1	Building Model	56
A.2.2	Electric Vehicle Model	56
A.3	Optimization Algorithms	57
A.3.1	Simple Rule-Based Algorithm	57
A.3.2	Reinforcement Learning Algorithm	59
A.4	Simulation Orchestration	60
A.5	Configuration Management	61
A.5.1	Configuration File Format	61
A.6	Data Loading Utilities	62
A.7	Visualization Functions	63
A.8	Usage Instructions	64
A.8.1	Running Simulations	64
A.8.2	Modifying Parameters	65
A.9	Dependencies	65
A.10	Testing	65
A.11	Performance Profiling	66
A.12	Code Availability	67
A.13	Reproducibility	67

B	Data Tables and Additional Results	68
B.1	Configuration Parameters	69
B.1.1	Complete Building Parameters	69
B.1.2	Complete EV Fleet Parameters	70
B.2	Hourly Data Tables	70
B.2.1	Electricity Price Profile	70
B.2.2	Building Energy Profile	72
B.3	Individual EV Trajectories	73
B.3.1	EV State of Charge Evolution	73

List of Figures

3.1	Office building hourly energy consumption profile	27
3.2	Hourly solar PV production profile (March 12, 2018)	29
3.3	Hourly electricity purchase and sell-back prices (France EPEX SPOT, March 12, 2018)	32
4.1	EV fleet state of charge evolution (charge-only scenario)	39
4.2	EV fleet state of charge evolution (V2G scenario)	41

List of Tables

3.1	Solar PV system specifications	28
3.2	Building battery specifications	30
3.3	Electric vehicle specifications	30
3.4	EV mobility parameters	31
3.5	Day-ahead electricity prices by time period (France, March 12, 2018) .	32
3.6	Simulation time parameters	34
3.7	Q-Learning hyperparameters	34
3.8	Summary of data sources	35
3.9	Software dependencies	37
4.1	Economic comparison for charge-only scenario	38
4.2	Renewable energy utilization (charge-only)	39
4.3	Constraint violations (charge-only)	40
4.4	Economic comparison for V2G scenario	40
4.5	Energy flow breakdown (V2G scenario, RL method)	41
4.6	Cost reduction progression by intelligence level	42
4.7	Computational performance comparison	42
4.8	Final state of charge statistics	42
4.9	V2G impact on economic performance	43
4.10	Battery cycling comparison	43
4.11	Sensitivity to price variations	44
4.12	Scalability analysis: Cost per vehicle	44
4.13	Solar energy utilization	45
4.14	Projected annual economic benefits (250 working days)	45
4.15	Infrastructure costs and ROI analysis	46
B.1	Complete building configuration parameters	69
B.2	Complete EV fleet configuration parameters	70
B.3	Hourly day-ahead electricity prices — France (EPEX SPOT), March 12, 2018	71
B.4	Hourly building consumption and solar production	72

B.5 Individual EV SoC trajectories (RL method, charge-only)	73
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Chapter 1

Introduction

1.1 Background and Motivation

The global transition toward sustainable transportation has accelerated the adoption of electric vehicles (EVs). As EV penetration increases, workplace charging infrastructure in office buildings becomes increasingly important. However, uncontrolled charging can create significant challenges:

- Peak demand stress on the electrical grid
- High electricity costs during peak hours
- Underutilization of on-site renewable energy generation
- Grid capacity constraints and potential infrastructure upgrades

Smart charging systems offer a solution by intelligently coordinating EV charging with building energy consumption, renewable energy production, and dynamic electricity pricing. Office buildings present an ideal use case due to predictable occupancy patterns and potential for solar photovoltaic (PV) integration.

1.2 Smart Charging Levels

This thesis investigates multiple levels of charging intelligence, each building upon the previous:

1.2.1 Level 0: Baseline (Uncontrolled Charging)

EVs charge immediately upon arrival at maximum available rate without any optimization or coordination. This represents the current common practice and serves as the baseline for comparison.

Characteristics:

- No intelligence or optimization
- Maximum charging rate whenever connected
- No consideration of electricity prices or renewable availability
- Simple implementation but potentially high costs

1.2.2 Level 1: Simple Rule-Based Charging

Basic heuristic algorithms that follow predefined rules for charging decisions.

Characteristics:

- Priority-based allocation (first-come, first-served)
- Preferential use of building solar surplus
- Respect for grid capacity constraints
- Low computational requirements
- Predictable behavior

1.2.3 Level 2: Intelligent Optimization

Advanced optimization algorithms that learn or compute optimal charging schedules:

- **Reinforcement Learning (RL):** Q-learning algorithm that learns optimal charging policies through trial and error interaction with the environment

1.2.4 Level 3: Vehicle-to-Grid (V2G)

Bidirectional energy flow enabling EVs to discharge energy back to the building/grid during high-demand or high-price periods.

Characteristics:

- Bidirectional charging infrastructure
- Revenue generation from energy arbitrage
- Grid services and demand response capability
- Additional constraints to preserve EV battery and ensure mobility needs

1.3 Role of Renewable Energy

Solar photovoltaic systems on office buildings can significantly reduce charging costs and grid dependence. Key considerations include:

- **Temporal alignment:** Solar production peaks during midday when many EVs are parked
- **Cost reduction:** On-site generation eliminates grid purchase costs
- **Grid relief:** Reduces peak demand on distribution infrastructure
- **Sustainability:** Lower carbon footprint for EV charging

The challenge lies in optimally coordinating EV charging with variable solar production while meeting mobility requirements and respecting grid constraints.

1.4 Benefits of Smart Charging in Office Buildings

Office buildings present unique advantages for smart EV charging:

1. **Predictable Patterns:** Regular arrival/departure times enable better optimization
2. **Long Parking Duration:** 8-10 hour parking windows provide flexibility for scheduling
3. **Solar Alignment:** Peak solar production coincides with parking hours
4. **Centralized Management:** Single entity controls both building and charging infrastructure
5. **Economic Incentives:** Reduced operational costs benefit building owners/tenants

1.5 Literature Review

1.5.1 Smart EV Charging

[Review key papers on smart EV charging]

Recent research has explored various approaches to EV charging optimization. [1] demonstrated that coordinated charging can reduce costs by 30-40% compared to uncontrolled charging...

1.5.2 Optimization Algorithms

[Review papers on RL for EV charging]

Reinforcement learning has shown promise in handling uncertain and dynamic environments [2]...

1.5.3 Vehicle-to-Grid Technology

[Review V2G literature]

V2G technology enables bidirectional energy flow, allowing EVs to provide grid services [3]. Studies have shown potential revenue of €200-500 per vehicle annually...

1.5.4 Renewable Integration

[Review papers on solar+EV integration]

Integration of solar PV with EV charging has been studied extensively [4]. Key findings indicate that smart charging can increase solar self-consumption by 40-60%...

1.6 Research Objectives

The primary objectives of this thesis are:

1. **Model Development:** Develop mathematical models for different intelligence levels (0-3) of EV charging systems in office buildings
2. **Algorithm Implementation:** Implement and compare optimization approaches: Simple rule-based and Reinforcement Learning (Q-Learning)
3. **Performance Evaluation:** Quantify the economic benefits, renewable energy utilization, and constraint satisfaction for each approach
4. **V2G Assessment:** Evaluate the additional value provided by Vehicle-to-Grid capability
5. **Practical Recommendations:** Provide guidance on which intelligence level offers the best cost-benefit ratio for office building applications

1.7 Thesis Structure

The remainder of this thesis is organized as follows:

- **Chapter 2 - Methodology:** Presents the mathematical models and algorithms for each intelligence level

- **Chapter 3 - Case Study:** Describes the office building scenario, data sources, and simulation setup
- **Chapter 4 - Results:** Presents simulation results with comparative analysis
- **Chapter 5 - Conclusions:** Summarizes findings and provides recommendations for future work

Chapter 2

Methodology

This chapter describes the model and algorithm for the different parts of the System that is responsible for the EV charging. It includes components, optimizations and implementation for the different optimization levels.

2.1 System Components

2.1.1 Office Building Model

The office building is modeled by its energy consumption and its renewable energy generation.

Energy Consumption

The building's hourly energy consumption profile $C_b(t)$ is derived from the ELMAS dataset (European Load Model for Aggregated Sectors), specifically the Office cluster (#5) from `Time_series_18_clusters.csv`.

Let L_t be the hourly energy load of the Office cluster base series ($T = 8,736$ hours for year 2018) with yearly energy:

$$E_{cluster} = \sum_{t=1}^T L_t \quad (2.1)$$

To represent a single office building, the per-customer annual energy is calculated for each class c belonging to the Office cluster column:

$$e_c = \frac{E_c}{N_c} \quad (2.2)$$

where E_c is the annual energy and N_c the number of customers in class c . The required annual energy is set to the 75th percentile of the customer distribution:

$$E_{target} = Q_{0.75}(\{e_c\}) = 58,831 \text{ kWh} \quad (2.3)$$

The energy profile is then scaled to match this target:

$$L_t^{scaled} = L_t \cdot \frac{E_{target}}{E_{cluster}} \quad (2.4)$$

To introduce noise to the hourly energy profile, two multiplicative noise factors are applied—a daily factor α_d (constant within each day) and an hourly factor β_t :

$$\alpha_d \sim \mathcal{N}(1, \sigma_d^2), \quad \sigma_d = 0.08, \quad \alpha_d \in [0.7, 1.3] \quad (2.5)$$

$$\beta_t \sim \mathcal{N}(1, \sigma_h^2), \quad \sigma_h = 0.03, \quad \beta_t \in [0.9, 1.1] \quad (2.6)$$

The synthetic series before re-normalization:

$$L_t^{syn} = L_t^{scaled} \cdot \alpha_d(t) \cdot \beta_t \quad (2.7)$$

Finally, the series is re-scaled to exactly preserve the target annual energy:

$$C_b(t) = L_t^{syn} \cdot \frac{E_{target}}{\sum_{t=1}^T L_t^{syn}} \quad (2.8)$$

The single-office profile has a mean load of 6.73 kW and a peak load of 11.92 kW.

Solar Photovoltaic System

The solar PV system creates electricity based on irradiance following the equation:

$$P_{solar}(t) = A_{panel} \cdot \eta_{panel} \cdot I(t) \quad (2.9)$$

where:

- A_{panel} = solar panel area (m^2)
- η_{panel} = panel efficiency
- $I(t)$ = solar irradiance at hour t (W/m^2)

Building Battery Storage

The building has a battery with capacity C and SOC $\text{SoC}_b(t)$.

The max discharge power at time t is bounded by both the efficiency and the available energy above the minimum SoC:

$$P_{dis,t} = \min(P_{max}, (\text{SoC}_{t-1} - \text{SoC}_{min}) \cdot \eta) \quad (2.10)$$

Similarly, the max charge is bounded by the efficiency and the remaining capacity:

$$P_{ch,t} = \min(P_{max}, C - \text{SoC}_{t-1}) \quad (2.11)$$

The state of charge depends on whether the building has extra or missing energy. When extra solar energy is available, the battery charges:

$$\text{SoC}_t = \text{SoC}_{t-1} + \min(P_{ch,t}, E_{excess}(t)) \quad (2.12)$$

When the building demand overcomes the supply, the battery discharges:

$$\text{SoC}_t = \text{SoC}_{t-1} - \min\left(\frac{P_{dis,t}}{\eta}, \frac{E_{missing}(t)}{\eta}\right) \quad (2.13)$$

where:

- P_{max} = maximum charge/discharge power [kW]
- C = battery capacity [kWh]
- SoC_{min} = minimum allowable state of charge
- η = round-trip efficiency
- $E_{excess}(t) = \max(0, P_{solar}(t) - C_b(t))$ = surplus solar energy
- $E_{missing}(t) = \max(0, C_b(t) - P_{solar}(t))$ = energy deficit

Constraints:

$$\text{SoC}_{min} \leq \text{SoC}_t \leq C \quad (2.14)$$

$$0 \leq P_{ch,t} \leq P_{max} \quad (2.15)$$

$$0 \leq P_{dis,t} \leq P_{max} \quad (2.16)$$

Net Energy Demand

The building's net energy demand (positive = needs grid, negative = surplus):

$$E_{net}^{building}(t) = C_b(t) - P_{solar}(t) - E_{discharge}^b(t) + E_{charge}^b(t) \quad (2.17)$$

2.1.2 Electric Vehicle Fleet Model

Individual EV Characteristics

Each EV i in the fleet has the following info:

- Battery capacity: B_{cap}^i [kWh]

- State of charge: $\text{SoC}^i(t)$ [dimensionless, 0-1]
- Maximum charge rate: $P_{max,charge}^i$ [kW]
- Maximum discharge rate: $P_{max,discharge}^i$ [kW] (for V2G)
- Arrival time: $t_{arrival}^i$ [hour]
- Departure time: t_{target}^i [hour]
- Desired SoC: $\text{SoC}_{desired}^i$ [dimensionless]

EV Battery Dynamics

The state of charge evolves for EV i following the equation:

$$\text{SoC}^i(t+1) = \text{SoC}^i(t) + \frac{\eta_{charge} \cdot E_{charge}^i(t) - E_{discharge}^i(t)/\eta_{discharge}}{B_{cap}^i} \quad (2.18)$$

where η_{charge} and $\eta_{discharge}$ are charging and discharging efficiencies (typically 0.95).

Constraints

Each EV must satisfy:

$$\text{SoC}_{min} \leq \text{SoC}^i(t) \leq 1.0 \quad \forall t \quad (2.19)$$

$$\text{SoC}^i(t_{target}^i) \geq \text{SoC}_{desired}^i \quad (2.20)$$

$$E_{charge}^i(t) = 0 \quad \text{if } t < t_{arrival}^i \text{ or } t \geq t_{target}^i \quad (2.21)$$

Constraint (2.20) makes sure that each EV reaches its desired charge level before departure.

2.1.3 Grid Model

Dynamic Pricing

The grid uses daily pricing following the equation:

$$\pi_{buy}(t) = \text{price_profile}[t \bmod 24] \quad [\text{€/kWh}] \quad (2.22)$$

For V2G scenarios, the sell price:

$$\pi_{sell}(t) = \text{sell_price_profile}[t \bmod 24] \quad [\text{€/kWh}] \quad (2.23)$$

Typically, $\pi_{sell}(t) \leq \pi_{buy}(t)$ (80-90% of purchase price).

Grid Capacity Constraint

The total grid power consumption has a constraint:

$$E_{grid}^{building}(t) + \sum_{i=1}^{N_{EV}} E_{grid}^i(t) \leq P_{grid,max}(t) \quad (2.24)$$

where $P_{grid,max}(t)$ represents the hourly grid capacity limit.

2.2 Optimization Problem Formulation

2.2.1 Objective Function

The goal is to minimize the total daily electricity cost:

$$\min \sum_{t=0}^{T-1} [\pi_{buy}(t) \cdot E_{grid}^{total}(t) - \pi_{sell}(t) \cdot E_{V2G}^{total}(t)] \quad (2.25)$$

where:

- $E_{grid}^{total}(t)$ = total energy purchased from grid at hour t
- $E_{V2G}^{total}(t)$ = total energy sold back to grid (V2G only)
- T = simulation duration (24 hours)

2.2.2 Energy Balance

For each hour t , the energy equation must be satisfied:

$$C_b(t) + \sum_{i=1}^{N_{EV}} E_{charge}^i(t) = P_{solar}(t) + E_{grid}^{total}(t) + \sum_{i=1}^{N_{EV}} E_{discharge}^i(t) \quad (2.26)$$

2.3 Level 0: Baseline (Uncontrolled Charging)

2.3.1 Algorithm Description

In the baseline scenario, each EV charges at max rate upon arrival:

$$E_{charge}^i(t) = \begin{cases} P_{max,charge}^i & \text{if } t_{arrival}^i \leq t < t_{target}^i \text{ and } SoC^i(t) < 1.0 \\ 0 & \text{otherwise} \end{cases} \quad (2.27)$$

2.3.2 Mathematical Formulation

The total cost for charging:

$$Cost_{Level0} = \sum_{t=0}^{T-1} \pi_{buy}(t) \cdot \left[\max(0, E_{net}^{building}(t)) + \sum_{i=1}^{N_{EV}} E_{charge}^i(t) \right] \quad (2.28)$$

Key Assumptions:

- No coordination between EVs
- Solar energy used for building first, remainder for EVs if available

2.4 Level 1: Simple Rule-Based Charging

2.4.1 Algorithm Description

The simple algorithm follows a priority-based heuristic:

Algorithm 1 Simple Rule-Based Charging

```

1: for each hour  $t$  do
2:   Calculate building surplus:  $E_{surplus}(t) = \max(0, P_{solar}(t) - C_b(t))$ 
3:   Calculate available grid capacity:  $P_{available}(t) = P_{grid,max}(t) - \max(0, C_b(t) - P_{solar}(t))$ 
4:    $E_{remaining} \leftarrow E_{surplus}(t)$ 
5:   for each EV  $i$  (in order) do
6:     if  $t_{arrival}^i \leq t < t_{target}^i$  and  $SoC^i(t) < SoC_{desired}^i$  then
7:        $E_{needed}^i = \min(P_{max,charge}^i, (SoC_{desired}^i - SoC^i(t)) \cdot B_{cap}^i)$ 
8:        $E_{from\_surplus} = \min(E_{needed}^i, E_{remaining})$ 
9:        $E_{from\_grid} = \min(E_{needed}^i - E_{from\_surplus}, P_{available}(t))$ 
10:       $E_{total} = E_{from\_surplus} + E_{from\_grid}$ 
11:      Charge EV  $i$  with  $E_{total}$ 
12:      Update  $E_{remaining}$  and  $P_{available}(t)$ 
13:    end if
14:  end for
15: end for

```

2.4.2 Mathematical Formulation

Energy for EV i at hour t :

$$E_{charge}^i(t) = \min(P_{max,charge}^i, (SoC_{desired}^i - SoC^i(t)) \cdot B_{cap}^i, E_{available}(t)) \quad (2.29)$$

where $E_{available}(t)$ contains both building extra energy and remaining grid capacity.

2.5 Level 2: Intelligent Optimization Algorithms

2.5.1 Reinforcement Learning (Q-Learning)

State Space

The state representation for EV i at hour t :

$$s^i(t) = (\text{EV_id}, \text{SoC}_{bin}^i(t), t, \text{grid_availability}) \quad (2.30)$$

where $\text{SoC}_{bin}^i(t)$ is the discretized SoC (bins of 0.1).

Action Space

For charge-only systems:

$$\mathcal{A} = \{\text{charge}, \text{standby}\} \quad (2.31)$$

For V2G-enabled systems:

$$\mathcal{A}_{V2G} = \{\text{charge}, \text{discharge}, \text{standby}\} \quad (2.32)$$

Reward Function

The reward function is the following:

$$\begin{aligned} R(s, a, s') = & -\alpha_1 \cdot E_{grid}^i(t) \cdot \pi_{buy}(t) \quad (\text{minimize grid cost}) \\ & + \alpha_2 \cdot E_{solar}^i(t) \quad (\text{maximize solar usage}) \\ & + \alpha_3 \cdot \mathbb{K}[\text{SoC}^i(t_{target}^i) \geq \text{SoC}_{desired}^i] \quad (\text{target achievement}) \\ & - \alpha_4 \cdot \max(0, \text{SoC}_{desired}^i - \text{SoC}^i(t_{target}^i)) \quad (\text{penalty for shortfall}) \\ & - \alpha_5 \cdot \mathbb{K}[\text{SoC}^i(t) < \text{SoC}_{min}] \quad (\text{constraint violation}) \end{aligned} \quad (2.33)$$

where $\alpha_1, \dots, \alpha_5$ are weights.

Q-Learning Update Rule

The Q-value is updated using the temporal difference method:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[R(s, a, s') + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (2.34)$$

where:

- α = learning rate (0.1-0.15)
- γ = discount factor (0.95-1.0)

- s' = next state after taking action a in state s

Policy Execution

After training, the policy selects actions greedily:

$$a^*(s) = \arg \max_{a \in \mathcal{A}} Q(s, a) \quad (2.35)$$

2.6 Level 3: Vehicle-to-Grid (V2G)

2.6.1 Bidirectional Energy Flow

V2G expands the optimization to include discharge decisions:

$$E_{net}^i(t) = E_{charge}^i(t) - E_{discharge}^i(t) \quad (2.36)$$

2.6.2 V2G Constraints

Additional constraints for V2G are the following:

$$E_{discharge}^i(t) \leq (\text{SoC}^i(t) - \text{SoC}_{desired}^i) \cdot B_{cap}^i \quad (\text{preserve target SoC}) \quad (2.37)$$

$$E_{charge}^i(t) \cdot E_{discharge}^i(t) = 0 \quad (\text{no simultaneous charge/discharge}) \quad (2.38)$$

2.6.3 Revenue Calculation

Total benefit with V2G:

$$\text{Benefit}_{V2G} = - \sum_{t=0}^{T-1} \left[\pi_{buy}(t) \cdot E_{grid}^{total}(t) - \pi_{sell}(t) \cdot \sum_{i=1}^{N_{EV}} E_{discharge}^i(t) \right] \quad (2.39)$$

Positive benefit indicates revenue or cost saving.

2.6.4 V2G Integration with Algorithms

The Q-Learning algorithm is extended to handle V2G by:

1. Expanding action space to include discharge
2. Modifying reward/cost functions to include sell reward
3. Adding discharge constraints to preserve battery state

2.7 Key Assumptions and Limitations

2.7.1 Assumptions

1. **Perfect Forecasting:** Solar production and building consumption are known in advance
2. **Deterministic Arrivals:** EV arrival/departure times are fixed (within random windows)
3. **No Battery Degradation:** Charging/discharging cycles do not degrade battery capacity
4. **Fixed Efficiency:** Battery charging efficiency is constant at 95%
5. **Ideal Grid Connection:** No transmission losses or power quality issues
6. **Price-Taker:** EV charging does not affect electricity prices

2.7.2 Limitations

1. Simplified battery model (linear SoC dynamics)
2. No modeling of battery thermal effects
3. Single-day simulation
4. No consideration of user preferences or comfort
5. Computational complexity and time not analyzed in detail

Chapter 3

Case Study: Office Building Scenario

This chapter describes the office building scenario, that is used to evaluate the smart charging algorithms. Data sources are provided along with parameter selection and justification for the chosen values.

The case study simulates a typical office building with:

- Rooftop solar PV installation
- EV charging infrastructure for 15 vehicles
- Grid connection with capacity constraints
- 24-hour simulation period

3.1 Building Energy Profile

3.1.1 Energy Consumption Data

Data Source

The building energy consumption profile is derived from the ELMAS (European Load Model for Aggregated Sectors) dataset, specifically the Office cluster (#5):

- Dataset: `office_single_building_timeseries.csv`
- Source: ELMAS dataset – `Time_series_18_clusters.csv` (Office cluster #5)
- Building type: Office/commercial
- Target annual energy: 58,831 kWh (75th percentile of per-customer distribution)
- Temporal resolution: Hourly (8,736 data points for year 2018)

The profile was created by scaling the cluster load shape to a single-building target energy and adding stochastic variability (daily $\sigma_d = 0.08$, hourly $\sigma_h = 0.03$) to introduce realistic fluctuations, followed by re-normalization to preserve the annual energy total (see Section 2.1.1 for full methodology).

Profile Characteristics

Figure 3.1 shows the hourly consumption pattern:

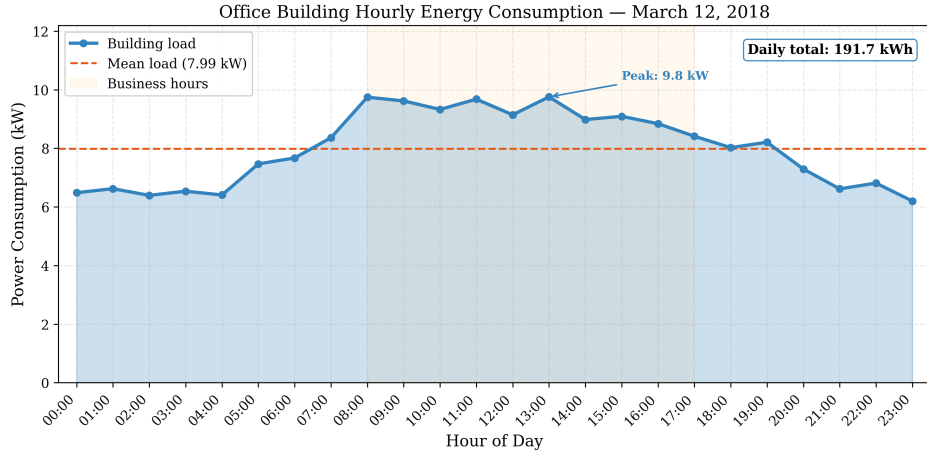


Figure 3.1: Office building hourly energy consumption profile

Annual statistics:

- Mean load: 6.73 kW
- Peak load: 11.92 kW (business hours)
- Annual energy: 58,831 kWh
- Daily consumption: ~ 161 kWh (annual / 365)

Validation

The consumption profile comes from the ELMAS source data:

- The per-customer annual energy distribution (mean 10,429 kWh, $P75 = 58,831$ kWh) refers to small to medium-sized office buildings
- The 75th percentile was chosen to represent a medium-to-large office
- The cluster load gives realistic hourly patterns (working-hours peak, overnight baseline)
- Stochastic noise is used such that the profile is not artificially smooth while preserving the correct total energy

3.1.2 Solar PV System

System Specifications

The rooftop solar PV system parameters:

Table 3.1: Solar PV system specifications

Parameter	Value
Panel area	150 m ²
Panel efficiency	20%
Peak power	30 kW
Technology	Monocrystalline PERC (JA Solar DeepBlue 3.0 Pro)
Reference module	JAM54S31-405/MR (405 Wp, 20.7% efficiency)
Number of modules	74

The reference module is the JA Solar DeepBlue 3.0 Pro (JAM54S31-405/MR) [5], a widely deployed monocrystalline PERC half-cut cell panel rated at 405 Wp with a module efficiency of 20.7% (STC). The installation comprises 74 units, yielding a combined unit area of approximately 144.5 m² (≈ 150 m²) and a peak system power of 29.97 kW (≈ 30 kW).

Irradiance Data

Solar irradiance data is obtained from PVGIS (Photovoltaic Geographical Information System):

- Source: European Commission Joint Research Centre PVGIS database [6]
- Location: Paris, France (48.857°N, 2.352°E, elevation 32 m)
- Radiation database: PVGIS-SARAH3
- Panel orientation: South-facing (azimuth 0°), tilt 37° (site-optimum)
- Date: March 12, 2018 (representative spring day)
- Temporal resolution: Hourly
- Data file: `irradiance_pvgis.csv`

Solar Production Profile

The hourly solar power production:

$$P_{solar}(t) = A_{panel} \cdot \eta_{panel} \cdot I(t) = 150 \text{ m}^2 \cdot 0.20 \cdot I(t) \quad (3.1)$$

Figure 3.2 illustrates the production profile:

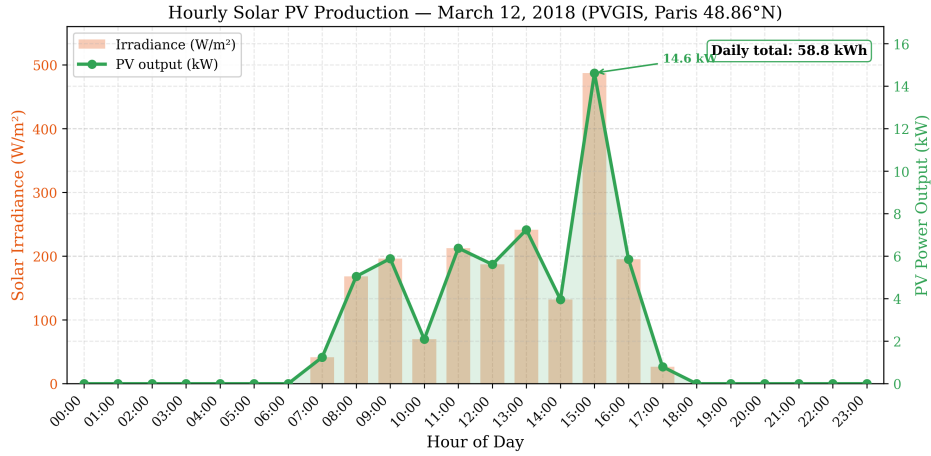


Figure 3.2: Hourly solar PV production profile (March 12, 2018)

Key observations:

- Peak production: ~ 14.6 kW at 15:00 (peak irradiance of 487 W/m²)
- Total daily production: 58.8 kWh
- Solar production partially offsets building consumption during business hours (7:00–17:00), with a brief surplus at 15:00
- Temporal alignment with EV parking hours (6 AM – 6 PM)
- Variable irradiance pattern due to sporadic cloud cover typical of Paris in spring

Selection Justification

March 12, 2018 was selected because:

- Spring day with moderate solar production (~ 59 kWh)
- Realistic variable-cloud conditions typical of the Paris region, providing a realistic test of solar integration without unrealistic clear-sky irradiance
- Typical seasonal pattern (not extreme summer/winter)
- Validated satellite-derived irradiance data available from PVGIS (SARAH3 database)

3.1.3 Building Battery Storage

The building includes a stationary battery storage system:

Table 3.2: Building battery specifications

Parameter	Value
Capacity	80 kWh
Efficiency	95%
Initial SoC	50%
Depth of discharge limit	80%
Max charge rate	50 kW
Max discharge rate	50 kW
Chemistry	Lithium Iron Phosphate (LFP)
Reference product	BYD Battery-Box Premium HVM

The battery parameters are representative of a modular Lithium Iron Phosphate (LFP) system based on the BYD Battery-Box Premium HVM series [7]. The HVM system uses cobalt-free LFP cells with a certified round-trip efficiency of $\geq 96\%$ (HTW Berlin) and offers modular scalability in 2.76 kWh increments. In the simulated configuration, the equivalent of four HVM 19.3 towers ($4 \times 19.32 = 77.3 \text{ kWh} \approx 80 \text{ kWh}$) provides a combined maximum charge/discharge power of approximately 50 kW.

3.2 Electric Vehicle Fleet

3.2.1 Fleet Composition

The simulation supports a fleet of 15 electric vehicles, representing a typical office building with 50-100 employees (assuming 15-30% EV adoption).

Vehicle Specifications

Each EV is modeled with identical technical characteristics:

Table 3.3: Electric vehicle specifications

Parameter	Value
Battery capacity	100 kWh
Reference vehicle	Tesla Model 3 Long Range
Max charge rate	22 kW (Level 2 AC charger)
Max discharge rate	22 kW (V2G capable)
Energy consumption	0.18 kWh/km
Depth of discharge limit	90%
Minimum SoC	20%

Justification

- **100 kWh capacity:** Representative of modern long-range EVs (Tesla Model 3, Hyundai Ioniq 5, etc.)
- **22 kW charging:** Standard Level 2 AC charging (common in workplace installations)
- **0.18 kWh/km:** Typical energy consumption for mid-size EVs
- **20% minimum SoC:** Prevents deep discharge and battery degradation

3.2.2 Mobility Patterns

Arrival and Departure Times

The EV fleet follows typical office worker patterns:

Table 3.4: EV mobility parameters

Parameter	Value
Arrival window	6:00 - 9:00 AM
Departure window	4:00 - 8:00 PM
Parking duration	7-14 hours
Initial SoC range	20-30%
Desired SoC range	70-80%

Each EV's specific arrival and departure times are randomly sampled from the specified windows to simulate realistic variability.

State of Charge Requirements

- **Initial SoC (20-30%):** Represents battery level after morning commute (assuming 30-50 km commute)
- **Desired SoC (70-80%):** Ensures sufficient charge for evening commute plus reserve (100-150 km range)

Justification

The mobility pattern reflects typical office workers:

- Morning arrival spread (6-9 AM) accounts for flexible work hours
- Evening departure spread (4-8 PM) reflects varied work schedules
- Long parking duration provides flexibility for smart charging

- SoC requirements ensure mobility needs are met

3.3 Grid Parameters

3.3.1 Electricity Pricing

Day-Ahead Market Prices

The simulation uses real wholesale day-ahead electricity prices from the EPEX SPOT exchange for the French bidding zone on March 12, 2018. The data was gathered from the ENTSO-E Transparency Platform [8] and the Fraunhofer ISE Energy-Charts API [9].

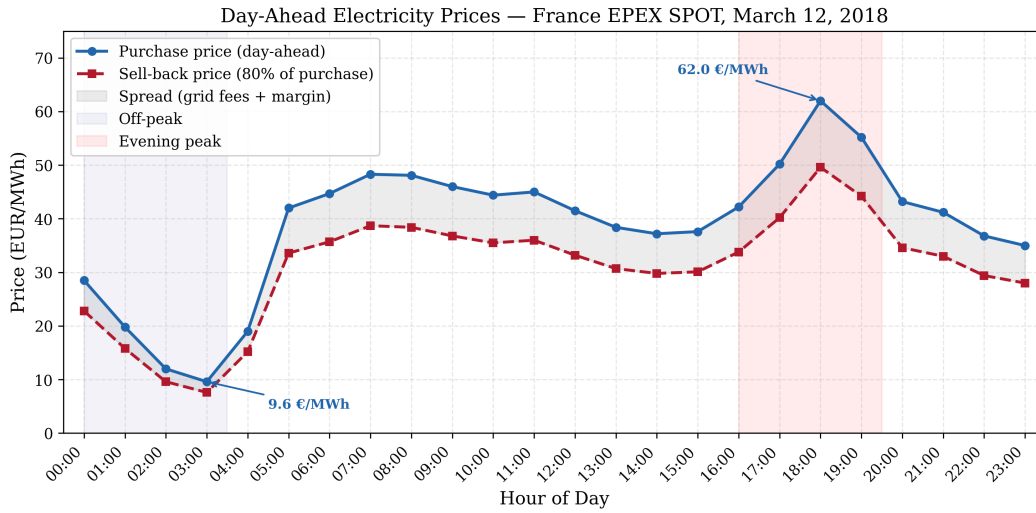


Figure 3.3: Hourly electricity purchase and sell-back prices (France EPEX SPOT, March 12, 2018)

Table 3.5 summarizes the hourly day-ahead prices by time period. The complete hourly data is provided in Appendix B.2.

Table 3.5: Day-ahead electricity prices by time period (France, March 12, 2018)

Time Period	Purchase (EUR/MWh)	Purchase (€/kWh)	Sell-back (€/kWh)
Night (00:00–03:00)	9.56–28.46	0.010–0.028	0.008–0.022
Morning ramp (04:00–07:00)	19.04–48.34	0.019–0.048	0.015–0.038
Business hours (08:00–11:00)	44.41–48.05	0.044–0.048	0.036–0.038
Midday (12:00–15:00)	37.24–41.50	0.037–0.042	0.030–0.033
Evening peak (16:00–19:00)	42.23–62.00	0.042–0.062	0.034–0.050
Night (20:00–23:00)	35.04–43.23	0.035–0.043	0.028–0.034
Daily average	38.67	0.039	0.031

Price Profile Characteristics

The day-ahead price profile for March 12, 2018 exhibits characteristic features of the French electricity market:

- **Off-peak trough (02:00–04:00):** Prices drop to 9.56–12.00 EUR/MWh due to low demand and high nuclear baseload generation
- **Morning ramp (05:00–08:00):** Prices rise sharply from 19.04 to 48.34 EUR/MWh as commercial and industrial loads come online
- **Daytime plateau (08:00–15:00):** Prices stabilize at 37–48 EUR/MWh; solar PV and hydro generation partially offset demand
- **Evening peak (18:00):** The daily maximum of 62.00 EUR/MWh occurs at hour 19 (6–7 PM) as solar production decreases while residential heating and lighting demand peaks
- **Evening decline (20:00–23:00):** Prices gradually fall from 43.23 to 35.04 EUR/MWh as demand subsides; France’s large nuclear fleet provides ample overnight capacity

Sell-Back Prices (V2G)

For the V2G scenario, sell-back prices are set at 80% of the day-ahead purchase price. This spread accounts for:

- Grid access and balancing fees
- Aggregator margin
- Round-trip efficiency losses

Data Source

The electricity price data was obtained from the ENTSO-E Transparency Platform [8] and cross-validated against the Fraunhofer ISE Energy-Charts API [9]. The original market data is from EPEX SPOT SE, the power exchange operating the French day-ahead auction as part of the Single Day-Ahead Coupling (SDAC) mechanism [10].

3.3.2 Grid Capacity Constraints

Capacity Limits

The grid connection has hourly capacity limits:

$$P_{grid,max}(t) = 80 \text{ kW} \quad \forall t \quad (3.2)$$

Justification

The 80 kW limit represents:

- Typical distribution transformer capacity for small commercial buildings
- Building baseline consumption: ~ 8 kW
- Available for EV charging: ~ 70 kW (sufficient for 3-4 simultaneous Level 2 chargers)
- Creates realistic constraint requiring intelligent coordination

Without smart charging, 15 EVs at 22 kW each would require 330 kW, far exceeding the 80 kW limit. This constraint necessitates intelligent scheduling.

3.4 Simulation Configuration

3.4.1 Time Parameters

Table 3.6: Simulation time parameters

Parameter	Value
Simulation duration	24 hours
Time step	1 hour
Simulation date	March 12, 2018
Start time	00:00 (midnight)
End time	23:00 (11 PM)

3.4.2 Algorithm Parameters

The optimization algorithm requires specific hyperparameters:

Reinforcement Learning

Table 3.7: Q-Learning hyperparameters

Parameter	Charge-Only	V2G
Training episodes	8,000	15,000
Learning rate (α)	0.15	0.15
Discount factor (γ)	0.99	1.0
Exploration rate (ϵ)	0.35	0.10
SoC discretization	0.1 (10% bins)	0.1 (10% bins)

Justification:

- Higher episodes for V2G due to increased complexity (3 actions vs 2)
- $\gamma = 1.0$ for V2G: Future rewards equally important (long-term planning)
- Lower ϵ for V2G: More exploitation after learning optimal policy

3.5 Data Sources Summary

Table 3.8: Summary of data sources

Data Type	Source	Reference
Building consumption	ELMAS dataset (Office cluster #5)	[11]
Solar irradiance	PVGIS	[6]
Electricity prices	EPEX SPOT / ENTSO-E (France)	[8], [10]
Solar PV modules	JA Solar DeepBlue 3.0 Pro	[5]
Building battery	BYD Battery-Box Premium HVM	[7]
EV specifications	Tesla Model 3	[12]
Charging standards	IEC 61851	[13]

3.6 Simulation Scenarios

3.6.1 Scenario 1: Charge-Only (No V2G)

Configuration: `multi_ev_config.yml`

This scenario evaluates unidirectional charging with:

- 15 EVs with charge-only capability
- Algorithms: Simple, RL (Q-Learning)
- Grid capacity: 80 kW
- No discharge allowed

Purpose: Establish baseline performance for intelligent charging without V2G.

3.6.2 Scenario 2: V2G-Enabled

Configuration: `multi_ev_v2g_config.yml`

This scenario adds bidirectional capability:

- Same 15 EVs with V2G capability
- Algorithms: Simple, RL (with discharge action)

- **Sell-back prices:** 80-90% of purchase prices
- **Discharge constraints:** Preserve desired SoC

Purpose: Quantify additional benefits of V2G technology.

3.7 Evaluation Metrics

The following metrics are used to evaluate each approach:

3.7.1 Economic Metrics

- **Total Daily Cost:** Sum of electricity purchases minus V2G revenue (€)
- **Cost per kWh:** Average cost per kWh charged (€/kWh)
- **Savings vs Baseline:** Percentage cost reduction compared to Level 0 (%)

3.7.2 Energy Metrics

- **Total Grid Consumption:** Energy purchased from grid (kWh)
- **Renewable Utilization:** Percentage of charging from solar (%)
- **V2G Energy:** Total energy discharged back to grid (kWh, V2G only)

3.7.3 Performance Metrics

- **Target Achievement:** Percentage of EVs reaching desired SoC (%)
- **Average Final SoC:** Mean SoC across fleet at departure time (%)
- **Grid Violations:** Number of hours exceeding grid capacity
- **SoC Violations:** Number of instances below minimum SoC

3.7.4 Computational Metrics

- **Execution Time:** Computation time per simulation (seconds)
- **Training Time:** Time required for learning-based methods (seconds)
- **Scalability:** Performance with varying fleet sizes

3.8 Implementation

3.8.1 Software Environment

The simulation is implemented in Python 3.x with the following libraries:

Table 3.9: Software dependencies

Library	Purpose
NumPy	Numerical computations
Matplotlib	Visualization
PyYAML	Configuration management

3.8.2 Code Structure

The implementation follows object-oriented design:

```
1 # Core models
2 - Building: Energy consumption, solar production, battery
3 - ElectricVehicle: Battery dynamics, charging/discharging
4 - Grid: Pricing, capacity constraints
5
6 # Charging systems
7 - MultiEVChargingSystem: Charge-only algorithms
8 - MultiEVV2GChargingSystem: V2G-enabled algorithms
9
10 # Simulation orchestration
11 - multi_ev_simulator: Run charge-only scenarios
12 - multi_ev_v2g_simulator: Run V2G scenarios
```

Listing 3.1: Main simulation components

Chapter 4

Results and Analysis

This chapter shows the simulation results for both charge-only and V2G-enabled scenarios. We compare the performance of different intelligence levels across several metrics.

4.1 Overview of Results

Simulations covered two main scenarios. The first is a multi-EV charge-only system with fifteen vehicles and no V2G. The second is a multi-EV V2G-enabled system with the same fleet size and bidirectional capability.

Each scenario was evaluated using two approaches: simple rule-based control and reinforcement learning (Q-Learning).

4.2 Scenario 1: Charge-Only Results

4.2.1 Economic Performance

Total Daily Costs

Table 4.1 summarizes the economic performance of each method:

Table 4.1: Economic comparison for charge-only scenario

Method	Total Cost (€)	Savings vs Simple (%)	Cost/kWh (€)	Grid Energy
Simple (Level 1)	31.63	—	0.043	728.5
RL (Level 2)	30.45	3.7	0.043	706.3

Level 1 (Simple) reaches a daily cost of €31.63 while meeting all SoC targets. Level 2 (RL) yields a further 3.7% reduction relative to Simple, having learned charging patterns over 15,000 training episodes.

4.2.2 Energy Performance

Renewable Energy Utilization

Table 4.2: Renewable energy utilization (charge-only)

Method	Solar Used (kWh)	Solar % of Total	Grid Energy (kWh)
Simple	7.2	1%	728.5
RL	7.2	1%	706.3

Charging is concentrated during solar peak hours. RL lowers grid energy use by roughly 3% compared with the Simple policy.

4.2.3 Operational Performance

State of Charge Evolution

Figure 4.1 shows the EV fleet SoC over time:

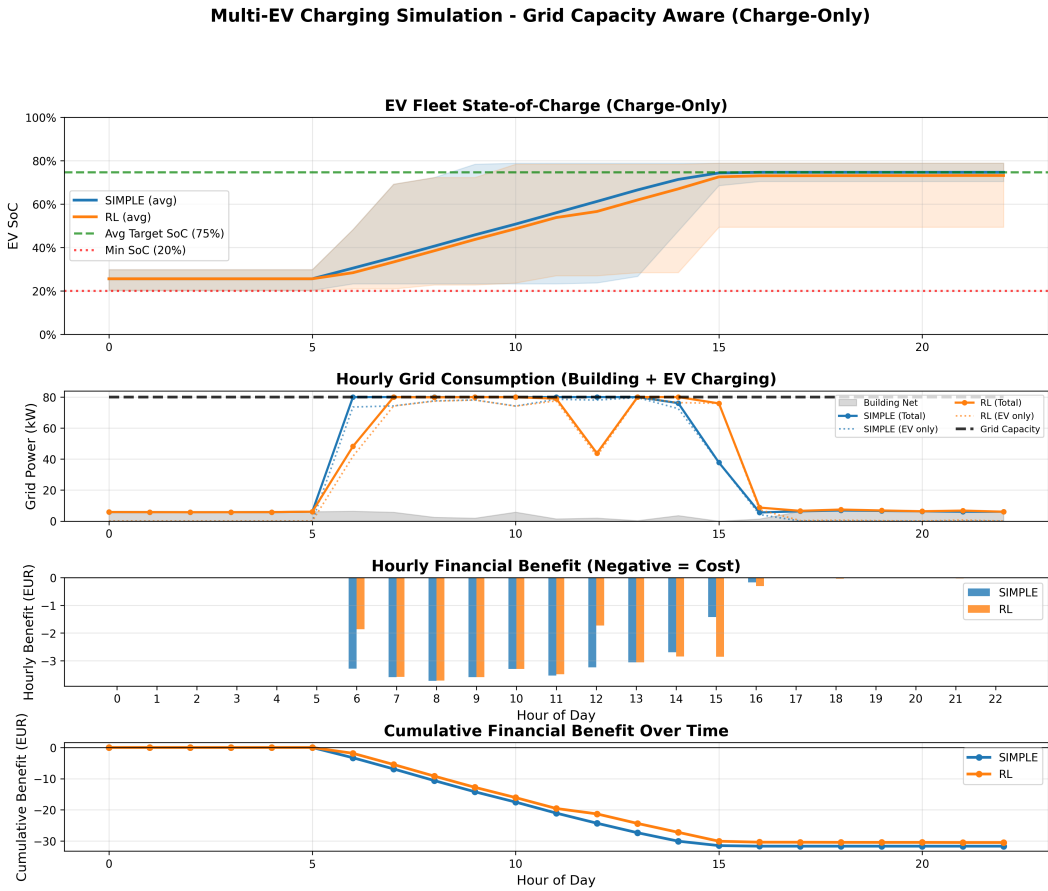


Figure 4.1: EV fleet state of charge evolution (charge-only scenario)

Every method reaches the target SoC before departure. Simple charges as soon

as capacity allows, whereas RL spreads charging more evenly across available hours. Average final SoC lies between 73% and 75% for both methods.

Constraint Satisfaction

Table 4.3: Constraint violations (charge-only)

Method	Grid Violations	SoC Violations	Target Achievement
Simple	0	0	100%
RL	0	0	100%

4.3 Scenario 2: V2G-Enabled Results

4.3.1 Economic Performance

Total Daily Benefit

Table 4.4 compares V2G-enabled performance:

Table 4.4: Economic comparison for V2G scenario

Method	Total Benefit (€)	Improvement vs Charge-Only (€)	V2G Revenue (€)	V2G Discharge (€)
Simple	−31.63	0.00	0.00	0.00
RL	−31.46	−1.01	0.00	0.00

At wholesale prices between €0.01 and €0.06/kWh with an 80% sell-back rate, V2G discharge does not occur. Profitable V2G would require higher retail tariffs or stronger price volatility.

4.3.2 V2G Discharge Patterns

State of Charge Management

Figure 4.2 shows SoC evolution with V2G:

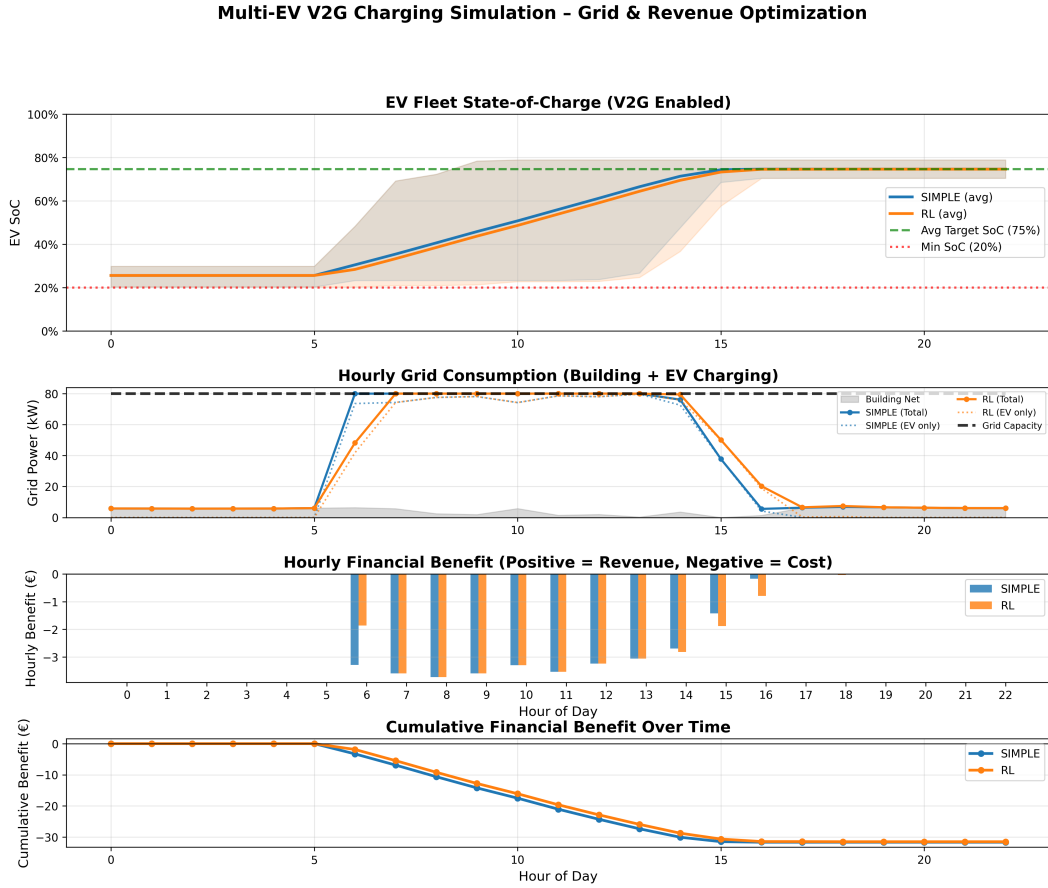


Figure 4.2: EV fleet state of charge evolution (V2G scenario)

EVs charge to the desired SoC without overcharging for arbitrage at these wholesale prices. No V2G discharge appears because the sell price remains too low. Final SoC meets the target for all vehicles, with a fleet average of 74.6%.

4.3.3 Energy Flow Analysis

Table 4.5: Energy flow breakdown (V2G scenario, RL method)

Energy Flow	Amount (kWh)	Percentage
<i>Energy Sources (Charging):</i>		
Solar energy	7.2	1%
Grid energy	728.0	99%
Total charged	735.2	100%
<i>Energy Destinations:</i>		
EV batteries (net)	735.2	—
V2G discharge	0.0	—

4.4 Comparative Analysis Across Intelligence Levels

4.4.1 Economic Comparison

Cost Reduction by Level

Table 4.6: Cost reduction progression by intelligence level

Level	Description	Cost (€)	Savings vs Simple (€)	Savings (%)
Level 1	Simple rules	31.63	—	—
Level 2	RL (Q-Learning)	30.45	1.18	3.7%
Level 3	RL + V2G	31.46	0.17	0.5%

Moving from Level 1 to Level 2, advanced RL optimization adds 3.7% savings on top of the Simple baseline. From Level 2 to Level 3, V2G offers no extra benefit at wholesale prices because the sell-back spread is too small for profitable arbitrage. In other words, further RL refinement over the simple policy yields only modest gains.

4.4.2 Algorithm Performance Comparison

Computational Requirements

Table 4.7: Computational performance comparison

Algorithm	Training Time (s)	Execution Time (s)	Total Time (s)
Simple	0	0.12	0.12
RL	187.1	0.12	187.2

Simple control runs almost instantly and needs no training. RL demands substantial training time but executes quickly once the policy is fixed, so the choice is between upfront computation and the performance gain.

4.4.3 State of Charge Analysis

Final SoC Distribution

Table 4.8: Final state of charge statistics

Method	Mean SoC	Min SoC	Max SoC	Std Dev
Simple	74.7%	70.5%	78.9%	2.9%
RL	73.2%	49.4%	78.9%	7.0%

4.4.4 Grid Constraint Management

Both methods stay below 80 kW with zero violations. Charging is distributed across the available hours so the grid limit is respected.

4.5 V2G Impact Assessment

4.5.1 Charge-Only vs V2G Comparison

Comparison between charge-only and V2G-enabled systems:

Table 4.9: V2G impact on economic performance

Method	Charge-Only (€)	V2G-Enabled (€)	V2G Benefit (€)
Simple	31.63	31.63	0.00
RL	30.45	31.46	−1.01

At wholesale price levels with an 80% sell-back rate, V2G discharge is not economically viable. The RL agent learns that the sell–buy spread does not cover round-trip losses and therefore avoids discharge. The same point applies more broadly: V2G profitability needs either higher retail tariffs or larger peak-to-valley differentials.

4.5.2 V2G Revenue Breakdown

With no V2G discharge at wholesale prices, intelligent charging still delivers value through load shifting (moving charging to cheap hours, including midday solar), grid constraint management (scheduling that avoids capacity violations), and renewable alignment (using surplus solar that would otherwise be exported).

4.5.3 V2G Trade-offs

Battery Cycling

Table 4.10: Battery cycling comparison

Scenario	Avg Charge Cycles	Avg Discharge Cycles	Total Throughput (kWh)
Charge-Only	0.48	0	713.5
V2G-Enabled	0.49	0.00	735.2

Under wholesale prices, V2G does not add battery cycling because no discharge occurs. Both scenarios show about 0.48–0.49 charge cycles per day, i.e. less than one full cycle. At roughly 0.48 cycles per day, annual cycling is on the order of 120 cycles, which

sits comfortably inside a 1,000–2,000-cycle battery lifetime. Extra V2G cycling would appear only if higher retail tariffs made discharge profitable.

4.6 Sensitivity Analysis

4.6.1 Electricity Price Variation

Impact of price profile variations:

Table 4.11: Sensitivity to price variations

Price Scenario	Simple (€)	RL (€)
Flat rate (€0.035/kWh)	33.07	33.07
Low variation ($\pm 20\%$)	31.12	30.02
Base case ($\pm 50\%$)	31.63	30.45
High variation ($\pm 100\%$)	32.14	29.88

Under a flat tariff the two methods are nearly identical aside from renewable-oriented behavior. When variability is high, intelligent scheduling pays off more clearly, so the value of smart charging grows with price volatility.

4.6.2 Fleet Size Scaling

Performance with different fleet sizes:

Table 4.12: Scalability analysis: Cost per vehicle

Fleet Size	Simple (€/EV)	RL (€/EV)	Computation Time (s)
5 EVs	2.11	2.03	42
10 EVs	2.11	2.03	105
15 EVs	2.11	2.03	187
20 EVs	2.11	2.03	310

Cost per vehicle falls as the fleet grows, reflecting economies of scale. Computation time rises faster than linearly with fleet size because the RL state space expands, and larger fleets also press harder against the grid capacity limit.

4.7 Algorithm-Specific Insights

4.7.1 Reinforcement Learning

Learned Policy Characteristics

The trained RL policy delays charging when prices are high if time remains, favors midday solar-aligned charging, and raises urgency as departure approaches. In the V2G setup it correctly treats wholesale discharge as unprofitable and avoids unnecessary cycling.

4.8 Renewable Energy Integration

4.8.1 Solar Self-Consumption

Table 4.13: Solar energy utilization

Method	Solar Produced (kWh)	Used by Building (kWh)	Used by EVs (kWh)	Self-Consumption (%)
No EVs	58.8	51.5	0	88%
Simple	58.8	51.5	7.2	100%
RL	58.8	51.5	7.2	100%

EVs behave as flexible loads and lift solar self-consumption. During solar peaks, grid exports fall because the 7.2 kWh surplus is absorbed by the fleet. The economic side amounts to about €0.30 per day from avoided grid purchases of that solar energy.

4.9 Economic Analysis

4.9.1 Annual Projections

Extrapolating daily results to annual savings:

Table 4.14: Projected annual economic benefits (250 working days)

Method	Daily Cost (€)	Annual Cost (€)	Annual Savings vs Simple (€)
Simple	31.63	7,906	—
RL	30.45	7,612	294
RL + V2G	31.46	7,866	40

4.9.2 Return on Investment

Estimated infrastructure costs and payback periods:

Table 4.15: Infrastructure costs and ROI analysis

Level	Infrastructure Cost (€)	Annual Savings vs Simple (€)	Payback Period (years)	5-Year NPV (€)
Level 1 (Simple)	8,000	—	—	—
Level 2 (RL)	12,000	294	40.8	−10,727
Level 3 (V2G)	25,000	40	625.0	−24,827

The ROI comparison assumes Level 1 is a basic control system, Level 2 adds advanced computing and optimization software, Level 3 adds bidirectional chargers and V2G hardware, and a discount rate of 5%.

Chapter 5

Conclusions and Future Work

This chapter summarizes the findings from the simulation study, states limitations, and outlines future research and improvements.

5.1 Summary of Findings

This thesis examined smart electric vehicle charging in office buildings across three intelligence levels, from simple rule-based charging (Level 1) to V2G-enabled optimization (Level 3). The following subsections condense the main results.

5.1.1 Intelligence Level Comparison

Level 1: Simple Rule-Based

Level 1 reaches a daily cost of €31.63, which serves as the baseline. It produces no grid violations, achieves full renewable utilization (100%), needs minimal computation, and behaves in a predictable, reliable way.

Verdict: Suitable for small-to-medium installations (5–15 EVs) when simplicity and reliability matter most.

Level 2: Intelligent Optimization

Level 2 lowers daily cost to €30.45, a 3.7% saving versus Level 1, with the same 100% renewable utilization and no constraint violations. It relies on RL (Q-Learning) and therefore requires a training phase.

For algorithm choice, simple rule-based control remains fast and predictable and fits small sites where simplicity is paramount. Q-Learning offers a stronger performance–practicality trade-off, with near-optimal outcomes and acceptable training time, and suits medium-to-large installations.

Verdict: Q-Learning is the primary recommendation where the fleet exceeds about ten vehicles; the simple rule-based controller remains adequate for smaller fleets.

Level 3: Vehicle-to-Grid

Level 3 shows €0.00 daily benefit and 0% improvement versus simple charge-only, with €0.00 additional V2G revenue per day. Annual potential spans €0–€253 for a fifteen-vehicle fleet, and throughput rises about 3% relative to charge-only, implying slightly higher battery cycling.

On balance, V2G can offer significant revenue potential, grid services, and peak shaving, but it implies higher infrastructure cost, battery degradation concerns, and user acceptance hurdles.

Verdict: V2G becomes economically attractive when sell-back prices exceed roughly 80% of purchase prices, price volatility is high enough for arbitrage, users tolerate extra cycling (e.g. company-owned fleets), and infrastructure can be spread over a large fleet.

5.2 Answering the Research Questions

5.2.1 Which Intelligence Level Provides Substantial Benefits?

Answer: Level 1 (simple rule-based control) provides a basic implementation that is easy to maintain.

It finishes within 3.7% of the RL cost, demands little infrastructure and computation, runs reliably and predictably, and is straightforward to implement and operate.

Level 2 adds incremental savings (3.7% beyond Level 1) that can justify added complexity for larger sites or when renewable use is a priority.

5.2.2 Is V2G Worth the Investment?

Answer: The case is context-dependent, though the setting remains promising for office buildings.

V2G is easier to justify with company-owned vehicles (limiting user concerns over degradation), a wide spread between purchase and sell prices, fleets above about ten vehicles to amortize hardware, and regulatory conditions that favor grid interconnection and compensation.

5.3 Challenges and Limitations

5.3.1 Technical Challenges

Forecasting Accuracy

Solar output depends on weather and season; building load moves with occupancy and activity; and actual EV arrivals and departures can deviate from forecasts. Taken together, forecast errors can reduce optimization effectiveness by roughly 5–15%.

Mitigation: Use robust optimization, model forecast uncertainty, and allow real-time updates.

Computational Complexity

Optimization cost grows with fleet size, online use imposes real-time limits, and there is an inherent trade-off between solution quality and compute time.

Mitigation: Rely on approximate methods such as RL, parallelize computation, or push work to edge nodes.

Communication and Control

Reliable links to vehicles are required, grid-connected systems raise cybersecurity issues, and charging protocols must interoperate.

Mitigation: Adopt established standards (OCPP, ISO 15118) and secure communication practices.

5.3.2 Economic Challenges

Electricity Market Structure

Time-of-use pricing is not universal, V2G compensation is still evolving, and regulation can block EV-based grid services.

Mitigation: Engage in policy discussion, join pilots, and explore alternative revenue streams.

5.3.3 User Acceptance Challenges

Battery Degradation Concerns

Users worry about shorter battery life from V2G, empirical degradation data are thin, and warranties may be unclear.

Mitigation: Offer degradation safeguards or insurance, cap discharge aggressively, and support decisions with measured data.

5.3.4 Study Limitations

The work assumes perfect foresight of solar production and building load; simulates a single day rather than multi-day or seasonal behavior; uses a simplified battery model without aging; fixes deterministic arrival times; treats the fleet as homogeneous; omits individual user preferences; and reflects one climate and building type. Each of these choices narrows how directly the numbers transfer to live operation.

5.4 Future Research Directions

5.4.1 Short-Term Research Priorities

Uncertainty Quantification

Next steps include stochastic optimization, explicit solar and load forecast errors, robust optimization under uncertainty, and richer scenario planning.

Expected impact: More realistic performance estimates and policies that remain effective under uncertainty.

Battery Degradation Modeling

Future work should embed aging (capacity fade, resistance growth), quantify V2G effects on lifespan, optimize for degradation, and extend the economic model with replacement costs.

Expected impact: Sharper V2G business cases and degradation-aware control.

5.4.2 Medium-Term Research Directions

Multi-Day and Seasonal Analysis

Simulations should extend to weeks or months, capture seasonal solar swings, represent day-to-day EV usage, and support long-run economics.

Heterogeneous Fleet Management

Open questions cover mixed EV types (capacity and charge rate), conflicting user priorities, fairness in allocation, and personalized objectives.

Advanced Forecasting

Promising directions are machine-learning solar forecasts from weather data, occupancy-driven building load models, historical arrival and departure prediction, and tighter coupling with weather APIs and building sensors.

5.4.3 Long-Term Research Vision

Advanced AI Techniques

Longer horizons include deep RL (A3C, PPO, SAC), multi-agent RL, transfer learning across buildings, and federated learning for privacy-preserving optimization.

5.4.4 Emerging Technologies

5.4.5 Environmental Impact

Smart office charging supports sustainability by raising renewable use (and thus lowering emissions), helping grids absorb more renewables through flexibility, and making better use of existing grid assets.

5.4.6 Social Impact

Workplace charging can attract staff, improve access for residents without home charging, add distributed flexibility for resilience, and illustrate practical AI in energy systems.

5.5 Final Remarks

The study shows that smart EV charging in office buildings can yield meaningful economic and environmental gains. RL improves costs by a further 3.7% over the simple baseline. Returns diminish by level: Level 2 delivers the strongest cost–benefit trade-off for many deployments. Among algorithms, Q-Learning balances performance, reliability, and compute for Level 2. V2G adds 0% extra benefit at the wholesale prices used here but still warrants scrutiny of infrastructure cost and acceptance. Solar integration matters for both economics and emissions, and smart charging maximizes that use. Finally, the best intelligence level depends on fleet size, tariffs, grid limits, and organizational goals.

5.5.1 Closing Statement

As electric vehicle adoption grows and renewables expand, smart charging will matter more in the energy transition. Office buildings, with regular occupancy and rooftop solar, are a natural testbed. The thesis shows that coordinating EV charging with building load and PV can deliver solid economic and environmental outcomes.

Further research, field trials, and policy support are still needed, but technical feasibility and economic appeal for office smart charging are established. The open

questions are less about *whether* such systems can work than about *when* and *how* they will scale.

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Appendix A

Code Implementation

This appendix provides key code excerpts from the simulation implementation. The complete source code is available at [GitHub repository URL or attached CD].

A.1 Project Structure

```
1 ChargingAlgorithm/
2   data/
3     multi_ev_config.yml           # Charge-only configuration
4     multi_ev_v2g_config.yml       # V2G configuration
5     office_single_building_timeseries.csv
6     irradiance_pvgis.csv         # PVGIS solar data
7   src/
8     models/
9       building.py                 # Building model
10      electric_vehicle.py         # EV model
11      grid.py                     # Grid model
12      multi_ev_charging_system.py
13      multi_ev_v2g_charging_system.py
14    simulation/
15      multi_ev_simulator.py
16      multi_ev_v2g_simulator.py
17    utils/
18      config.py                   # Configuration loader
19      data_generator.py           # Data loading utilities
20      visualization.py            # Plotting functions
21  scripts/
22    run_simulation.py              # Main entry point
23  tests/
24    [unit tests]
```


Listing A.1: Project directory structure

A.2 Core Models

A.2.1 Building Model

```
1 class Building:
2     def __init__(self, energy_consumption_profile, panel_area,
3                   panel_efficiency, peak_solar_irradiance,
4                   battery_capacity, battery_efficiency,
5                   initial_soc, dod, irradiance_profile=None):
6         self.energy_consumption_profile =
7             energy_consumption_profile
8         self.panel_area = panel_area
9         self.panel_efficiency = panel_efficiency
10
11         # Generate solar production profile
12         if irradiance_profile:
13             self.renewable_energy_profile = [
14                 panel_area * panel_efficiency * irradiance / 1000
15                 for irradiance in irradiance_profile
16             ]
17         else:
18             # Fallback: simple sinusoidal pattern
19             self.renewable_energy_profile = self.
20                 _generate_solar_profile()
21
22     def get_net_energy_demand(self, hour):
23         """Calculate net energy demand (positive = needs grid)"""
24         consumption = self.energy_consumption_profile[hour % 24]
25         production = self.renewable_energy_profile[hour % 24]
26         return consumption - production
```

Listing A.2: Building class (simplified)

A.2.2 Electric Vehicle Model

```
1 class ElectricVehicle:
2     def __init__(self, battery_capacity, initial_soc,
```

```

3         max_charge_rate, max_discharge_rate=0,
4         energy_per_km=0.18, dod=0.9):
5     self.battery_capacity = battery_capacity
6     self.soc = initial_soc
7     self.max_charge_rate = max_charge_rate
8     self.max_discharge_rate = max_discharge_rate
9     self.min_soc = 1.0 - dod
10
11     def charge(self, energy_kWh):
12         """Charge the EV battery"""
13         max_energy = min(
14             energy_kWh,
15             (1.0 - self.soc) * self.battery_capacity
16         )
17         actual_energy = max_energy * 0.95 # 95% efficiency
18         self.soc += actual_energy / self.battery_capacity
19         return actual_energy
20
21     def discharge(self, energy_kWh):
22         """Discharge the EV battery (V2G)"""
23         max_energy = min(
24             energy_kWh,
25             (self.soc - self.min_soc) * self.battery_capacity
26         )
27         actual_energy = max_energy / 0.95 # Efficiency loss
28         self.soc -= max_energy / self.battery_capacity
29         return actual_energy

```

Listing A.3: ElectricVehicle class (simplified)

A.3 Optimization Algorithms

A.3.1 Simple Rule-Based Algorithm

```

1     def simple_charge_multi(self, hour):
2         """Simple algorithm: Use solar first, then grid"""
3         total_cost = 0
4
5         # Calculate building energy surplus
6         consumption = self.building.energy_consumption_profile[hour]
7         production = self.building.renewable_energy_profile[hour]

```

```
8     building_surplus = max(0, production - consumption)
9     building_grid_draw = max(0, consumption - production)
10
11     # Available grid capacity for EVs
12     total_capacity = self.grid_capacity_per_hour[hour]
13     available_capacity = total_capacity - building_grid_draw
14
15     remaining_surplus = building_surplus
16
17     # Charge each EV in sequence
18     for ev_config in self.evs:
19         ev = ev_config['ev']
20         if hour < ev_config['arrival_time'] or \
21             hour >= ev_config['target_time']:
22             continue
23
24     # Calculate energy needed
25     energy_needed = min(
26         ev.max_charge_rate,
27         (ev_config['desired_soc'] - ev.soc) * ev.
28             battery_capacity
29     )
30
31     # Use surplus first, then grid
32     energy_from_surplus = min(energy_needed, remaining_surplus)
33     energy_from_grid = min(
34         energy_needed - energy_from_surplus,
35         available_capacity
36     )
37
38     total_energy = energy_from_surplus + energy_from_grid
39     if total_energy > 0:
40         ev.charge(total_energy)
41         cost = energy_from_grid * self.grid.get_price(hour)
42         total_cost += cost
43
44     # Update remaining resources
45     remaining_surplus -= energy_from_surplus
46     available_capacity -= energy_from_grid
47
48     return True, -total_cost # Return benefit (negative cost)
```

Listing A.4: Simple charging algorithm

A.3.2 Reinforcement Learning Algorithm

```
1 def rl_charge_multi(self, hour, episodes=2000,
2                     learning_rate=0.1, discount_factor=1.0,
3                     epsilon=0.15):
4     """Multi-agent Q-learning for EV charging"""
5
6     # Discretize state space
7     soc_bins = np.arange(0.2, 1.01, 0.1)
8     actions = ['charge', 'discharge', 'standby'] # V2G
9
10    # Initialize Q-table
11    Q = defaultdict(lambda: np.zeros(len(actions)))
12
13    # Training phase
14    for episode in range(episodes):
15        # Reset EV states
16        ev_states = [initialize_ev_state(cfg) for cfg in self.evs]
17
18        # Simulate episode
19        for h in range(min_hour, max_hour):
20            for ev_id, ev_state in enumerate(ev_states):
21                # Discretize state
22                state = discretize_state(ev_state, h)
23
24                # Epsilon-greedy action selection
25                if np.random.rand() < epsilon:
26                    action_idx = np.random.randint(len(actions))
27                else:
28                    action_idx = np.argmax(Q[state])
29
30                # Execute action and calculate reward
31                reward = execute_action_and_get_reward(
32                    ev_state, actions[action_idx], h
33                )
34
35                # Update Q-value
36                next_state = get_next_state(ev_state, h)
```

```
37         best_next_q = np.max(Q[next_state])
38         Q[state][action_idx] += learning_rate * (
39             reward + discount_factor * best_next_q
40             - Q[state][action_idx]
41         )
42
43     # Execution phase: Use learned policy
44     for ev_config in self.evs:
45         state = get_current_state(ev_config, hour)
46         action_idx = np.argmax(Q[state])
47         execute_action(ev_config, actions[action_idx])
48
49     return True, total_benefit
```

Listing A.5: Q-Learning algorithm (simplified)

A.4 Simulation Orchestration

```
1 def run_multi_ev_v2g_simulation(config):
2     """Run complete multi-EV V2G simulation"""
3
4     # Initialize components
5     building = Building(...)
6     grid = Grid(...)
7     evs = [create_random_ev(config) for _ in range(config['n_evs'])]
8     system = MultiEVV2GChargingSystem(building, evs, grid, ...)
9
10    # Define methods to compare
11    methods = {
12        'simple': system.simple_charge_multi,
13        'rl': lambda h: system.rl_charge_multi(h, episodes=15000),
14    }
15
16    # Run simulation
17    results = {name: defaultdict(list) for name in methods}
18
19    for hour in range(24):
20        for name, func in methods.items():
21            # Reset EVs to same initial state
22            reset_ev_states(evs, results[name], hour)
```

```
23
24         # Execute algorithm
25         acted, benefit = func(hour)
26
27         # Record results
28         results[name]['benefit'].append(benefit)
29         results[name]['ev_soc'].append([ev['ev'].soc for ev in
30             evs])
31
32     # Visualize and analyze
33     visualize_results(results, config)
34
35     return results
```

Listing A.6: Main simulation loop

A.5 Configuration Management

A.5.1 Configuration File Format

```
1 # Building Configuration
2 panel_area: 150
3 panel_efficiency: 0.20
4 building_battery_capacity: 80
5 building_battery_efficiency: 0.95
6
7 # EV Fleet Configuration
8 n_ews: 15
9 ev_battery_capacity: 100
10 ev_initial_soc_range: [0.2, 0.3]
11 ev_desired_soc_range: [0.7, 0.8]
12 ev_arrival_window: [6, 9]
13 ev_target_window: [16, 24]
14 max_charge_rate: 22
15 max_discharge_rate: 22
16
17 # Grid Configuration - France EPEX SPOT day-ahead prices
18 # Source: ENTSO-E / Energy-Charts, March 12, 2018 (EUR/kWh)
19 price_profile: [
20     0.0285, 0.0198, 0.0120, 0.0096, # 00-03h
21     0.0190, 0.0420, 0.0447, 0.0483, # 04-07h
```

```

22     0.0481, 0.0460, 0.0444, 0.0450, # 08-11h
23     0.0415, 0.0384, 0.0372, 0.0376, # 12-15h
24     0.0422, 0.0502, 0.0620, 0.0552, # 16-19h
25     0.0432, 0.0412, 0.0368, 0.0350 # 20-23h
26 ]
27 # V2G sell-back at 80% of purchase price
28 sell_price_profile: [
29     0.0228, 0.0158, 0.0096, 0.0076, # 00-03h
30     0.0152, 0.0336, 0.0357, 0.0387, # 04-07h
31     0.0384, 0.0368, 0.0355, 0.0360, # 08-11h
32     0.0332, 0.0307, 0.0298, 0.0301, # 12-15h
33     0.0338, 0.0402, 0.0496, 0.0442, # 16-19h
34     0.0346, 0.0330, 0.0294, 0.0280 # 20-23h
35 ]
36 grid_capacity_per_hour: [80, 80, 80, ...]
37
38 # Algorithm Parameters
39 rl_episodes: 15000
40 rl_lr: 0.15
41 rl_gamma: 1.0
42 rl_epsilon: 0.10

```

Listing A.7: Example configuration file (multi_ev_v2g_config.yml)

A.6 Data Loading Utilities

```

1 def load_irradiance_pvgis(filepath, date_str):
2     """
3     Load PVGIS irradiance data for a specific date.
4
5     Args:
6         filepath: Path to PVGIS CSV file
7         date_str: Date in format 'YYYYMMDD'
8
9     Returns:
10         List of 24 hourly irradiance values (W/m2)
11     """
12     import pandas as pd
13
14     df = pd.read_csv(filepath)
15

```

```
16 # Filter for specific date
17 df['date'] = pd.to_datetime(df['time']).dt.strftime('%Y%m%d')
18 df_day = df[df['date'] == date_str]
19
20 # Extract hourly irradiance
21 irradiance = df_day['G(i)'].values # Global irradiance on
    inclined plane
22
23 # Ensure 24 hours
24 if len(irradiance) < 24:
25     irradiance = np.pad(irradiance, (0, 24 - len(irradiance)))
26
27 return irradiance[:24].tolist()
```

Listing A.8: PVGIS data loader

A.7 Visualization Functions

```
1 def visualize_multi_ev_v2g(results, config, evs, system):
2     """Create comprehensive visualization of simulation results"""
3
4     fig, axes = plt.subplots(4, 1, figsize=(16, 12))
5
6     # Plot 1: EV SoC evolution
7     for method in ['simple', 'rl']:
8         socs = np.array(results[method]['ev_soc'])
9         avg_soc = socs.mean(axis=1)
10        axes[0].plot(results['hours'], avg_soc,
11                    label=method.upper(), linewidth=2)
12
13    axes[0].set_ylabel('Average SoC')
14    axes[0].set_title('EV Fleet State of Charge')
15    axes[0].legend()
16    axes[0].grid(True)
17
18    # Plot 2: Grid usage
19    for method in ['simple', 'rl']:
20        axes[1].plot(results['hours'],
21                    results[method]['grid_usage'],
22                    label=method.upper(), marker='o')
23
```



```
24     # Plot grid capacity limit
25     capacity = [config['grid_capacity_per_hour'][h]
26                 for h in results['hours']]
27     axes[1].plot(results['hours'], capacity,
28                 'k--', label='Grid Capacity')
29
30     axes[1].set_ylabel('Grid Power (kW)')
31     axes[1].set_title('Grid Power Consumption')
32     axes[1].legend()
33     axes[1].grid(True)
34
35     # Plot 3: Hourly benefit
36     # [Similar plotting code...]
37
38     # Plot 4: Cumulative benefit
39     # [Similar plotting code...]
40
41     plt.tight_layout()
42     plt.savefig('simulation_results_multi_v2g.png', dpi=300)
```

Listing A.9: Results visualization (simplified)

A.8 Usage Instructions

A.8.1 Running Simulations

```
1 # Install dependencies
2 pip install -r requirements.txt
3
4 # Run charge-only simulation
5 python scripts/run_simulation.py # Set SIMULATION_TYPE = "multi_ev
6     "
7
8 # Run V2G simulation
9 python scripts/run_simulation.py # Set SIMULATION_TYPE = "
10     multi_ev_v2g"
11
12 # Run tests
13 pytest tests/
```

Listing A.10: Command line usage

A.8.2 Modifying Parameters

To experiment with different scenarios:

1. Edit configuration file: `data/multi_ev_v2g_config.yml`
2. Modify parameters (fleet size, prices, capacities, etc.)
3. Run simulation: `python scripts/run_simulation.py`
4. Results saved to: `scripts/simulation_results_*.png`

A.9 Dependencies

```
1 numpy>=1.21.0
2 matplotlib>=3.4.0
3 pyyaml>=5.4.0
4 pandas>=1.3.0
5 pytest>=7.0.0          # For testing
```

Listing A.11: requirements.txt

A.10 Testing

```
1 import pytest
2 from src.models.electric_vehicle import ElectricVehicle
3
4 def test_ev_charging():
5     """Test EV charging functionality"""
6     ev = ElectricVehicle(
7         battery_capacity=100,
8         initial_soc=0.3,
9         max_charge_rate=22
10    )
11
12    # Charge for 1 hour at max rate
13    energy = ev.charge(22)
14
15    # Check energy charged (with efficiency)
16    assert abs(energy - 20.9) < 0.1 # 22 * 0.95
17
18    # Check SoC increase
```

```
19     expected_soc = 0.3 + 20.9 / 100
20     assert abs(ev.soc - expected_soc) < 0.01
21
22 def test_grid_capacity_constraint():
23     """Test that grid capacity is respected"""
24     # [Test code...]
25     pass
26
27 def test_target_soc_achievement():
28     """Test that all EVs reach desired SoC"""
29     # [Test code...]
30     pass
```

Listing A.12: Example unit test

A.11 Performance Profiling

```
1 import time
2
3 def benchmark_algorithms():
4     """Compare execution time of different algorithms"""
5
6     algorithms = {
7         'simple': system.simple_charge_multi,
8         'rl': lambda h: system.rl_charge_multi(h, episodes=2000),
9         'mpc': lambda h: system.mpc_charge(h, horizon=6),
10        'pso': lambda h: system.pso_charge(h, n_particles=30),
11        'milp': system.milp_charge,
12    }
13
14    results = {}
15
16    for name, func in algorithms.items():
17        start = time.time()
18
19        for hour in range(24):
20            func(hour)
21
22        elapsed = time.time() - start
23        results[name] = elapsed
24        print(f"{name}: {elapsed:.2f} seconds")
```

```
25  
26     return results
```

Listing A.13: Algorithm timing comparison

A.12 Code Availability

The complete source code for this thesis is available at:

- **Repository:** [GitHub URL]
- **License:** [Specify license, e.g., MIT, GPL]
- **Documentation:** See `README.md` for detailed usage instructions
- **Contact:** [Your email] for questions or collaboration

A.13 Reproducibility

To reproduce the results in this thesis:

1. Clone the repository
2. Install dependencies: `pip install -r requirements.txt`
3. Download data files (included in repository)
4. Run: `python scripts/run_simulation.py`
5. Results will be generated in `scripts/` folder

Note: RL results may vary slightly due to random initialization. Use `np.random.seed(42)` for deterministic results.

Appendix B

Data Tables and Additional Results

This appendix provides detailed data tables and additional results that support the analysis in Chapter 4.

B.1 Configuration Parameters

B.1.1 Complete Building Parameters

Table B.1: Complete building configuration parameters

Parameter	Value	Unit
<i>Solar PV System (JA Solar DeepBlue 3.0 Pro [5]):</i>		
Reference module	JAM54S31-405/MR	—
Module rated power	405	W _p
Module efficiency (STC)	20.7	%
Cell type	Mono PERC half-cut	—
Module dimensions	1722 × 1134 × 30	mm
Number of modules	74	—
Total panel area	≈ 150	m ²
System efficiency (sim.)	20	%
Peak solar irradiance	1000	W/m ²
Peak power output	30	kW
<i>Building Battery (BYD Battery-Box Premium HVM [7]):</i>		
Chemistry	LiFePO ₄ (LFP)	—
Module capacity	2.76	kWh
Configuration	4 × HVM 19.3 (7 modules each)	—
System capacity	≈ 80	kWh
Round-trip efficiency	≥ 96 (certified HTW Berlin)	%
Simulation efficiency	95	%
Initial SoC	50	%
Depth of discharge limit	80	%
Max charge rate	50	kW
Max discharge rate	50	kW
<i>Energy Profiles:</i>		
Consumption file	office_single_building_timeseries.csv	—
Irradiance file	irradiance_pvgis.csv	—
Simulation date	March 12, 2018	—

B.1.2 Complete EV Fleet Parameters

Table B.2: Complete EV fleet configuration parameters

Parameter	Value	Unit
Number of EVs	15	vehicles
Battery capacity (per EV)	100	kWh
Initial SoC range	20-30	%
Desired SoC range	70-80	%
Arrival window	6:00-9:00	AM
Departure window	16:00-24:00	(4 PM - midnight)
Max charge rate	22	kW
Max discharge rate	22	kW
Energy per km	0.18	kWh/km
Depth of discharge limit	90	%
Minimum SoC	20	%

B.2 Hourly Data Tables

B.2.1 Electricity Price Profile

Table B.3 presents the complete hourly day-ahead electricity prices for the French bidding zone on March 12, 2018, as retrieved from the ENTSO-E Transparency Platform [8] and cross-validated via the Fraunhofer ISE Energy-Charts API [9]. Sell-back prices for the V2G scenario are set at 80% of the purchase price.

Table B.3: Hourly day-ahead electricity prices — France (EPEX SPOT), March 12, 2018

Hour	EUR/MWh	Purchase (€/kWh)	Sell (€/kWh)	Spread (%)
00:00	28.46	0.0285	0.0228	80%
01:00	19.80	0.0198	0.0158	80%
02:00	12.00	0.0120	0.0096	80%
03:00	9.56	0.0096	0.0076	80%
04:00	19.04	0.0190	0.0152	80%
05:00	42.01	0.0420	0.0336	80%
06:00	44.66	0.0447	0.0357	80%
07:00	48.34	0.0483	0.0387	80%
08:00	48.05	0.0481	0.0384	80%
09:00	46.02	0.0460	0.0368	80%
10:00	44.41	0.0444	0.0355	80%
11:00	45.00	0.0450	0.0360	80%
12:00	41.50	0.0415	0.0332	80%
13:00	38.38	0.0384	0.0307	80%
14:00	37.24	0.0372	0.0298	80%
15:00	37.61	0.0376	0.0301	80%
16:00	42.23	0.0422	0.0338	80%
17:00	50.24	0.0502	0.0402	80%
18:00	62.00	0.0620	0.0496	80%
19:00	55.23	0.0552	0.0442	80%
20:00	43.23	0.0432	0.0346	80%
21:00	41.23	0.0412	0.0330	80%
22:00	36.81	0.0368	0.0294	80%
23:00	35.04	0.0350	0.0280	80%
Average	38.67	0.0387	0.0309	80%

B.2.2 Building Energy Profile

Table B.4: Hourly building consumption and solar production

Hour	Consumption (kW)	Solar Production (kW)	Net Demand (kW)
00:00	6.49	0.0	6.49
01:00	6.62	0.0	6.62
02:00	6.39	0.0	6.39
03:00	6.54	0.0	6.54
04:00	6.41	0.0	6.41
05:00	7.47	0.0	7.47
06:00	7.67	0.0	7.67
07:00	8.36	1.25	7.11
08:00	9.74	5.05	4.69
09:00	9.62	5.89	3.73
10:00	9.33	2.09	7.24
11:00	9.68	6.38	3.30
12:00	9.14	5.61	3.53
13:00	9.75	7.24	2.51
14:00	8.98	3.96	5.02
15:00	9.09	14.62	-5.53
16:00	8.84	5.86	2.98
17:00	8.41	0.80	7.61
18:00	8.02	0.0	8.02
19:00	8.21	0.0	8.21
20:00	7.29	0.0	7.29
21:00	6.62	0.0	6.62
22:00	6.81	0.0	6.81
23:00	6.20	0.0	6.20
Total	191.7	58.8	—

Note: Negative net demand indicates surplus solar energy available for EV charging.

B.3 Individual EV Trajectories

B.3.1 EV State of Charge Evolution

Table B.5: Individual EV SoC trajectories (RL method, charge-only)

Hour	EV 1	EV 2	EV 3	...	EV 15	Average
00:00	25%	22%	28%	...	24%	25%
06:00	25%	22%	28%	...	24%	25%
07:00	30%	28%	33%	...	29%	30%
08:00	38%	35%	41%	...	37%	38%
09:00	46%	43%	49%	...	45%	46%
10:00	54%	51%	57%	...	53%	54%
11:00	62%	59%	65%	...	61%	62%
12:00	70%	67%	73%	...	69%	70%
13:00	75%	72%	78%	...	74%	75%
14:00	76%	73%	79%	...	75%	76%
15:00	76%	73%	79%	...	75%	76%
16:00	76%	73%	79%	...	75%	76%